Agent-Computer Observation Interfaces Enable Dynamic Computer Use

Anonymous submission

Abstract

Computer-use (CU) agents commonly condition each decision on one screenshot and receive no audio input. This action-coupled observation scheme omits transient visual content and speech that occur between decisions. We treat the *observation interface*—when the environment is sampled, which modalities are captured, and how observations are represented and retained—as a system design dimension distinct from the underlying model and action space.

We introduce the **Agent-Computer Observation Interface (AOI)**, a perception layer that monitors the screen and system audio between decisions. AOI constructs a structured observation record from gated keyframes, volume-gated speech transcription, and model-generated visual narration that persists as text. It augments image-and-text CU models without retraining or changing their action spaces.

We evaluate AOI on **DynaCU-Bench**: 100 dynamic browser tasks across 10 families, plus a 50-task static control. Across eight CU models from 7B open-source models to frontier systems, AOI improves task success by 17–48 percentage points; Claude Sonnet 4.6 achieves a three-seed mean of 78.7% (3.1 pp standard deviation). On a 12-task audio subset, AOI-equipped Claude completes 12 tasks, compared with 3 for OpenAI Realtime and 0 for Gemini Live. Component analyses show that keyframes provide most of their value through visual narration and that the preferred observation channels depend on the underlying model.

1 Introduction

Most computer-use (CU) agents operate through a discrete observation–action loop: the model receives a screenshot, predicts an action, and receives another screenshot after execution (Xie et al. 2024; Zhou et al. 2023; He et al. 2024). Because observation occurs only at these action boundaries, transient content that disappears before the next screenshot never enters the trajectory. Screenshot-based agents also lack access to spoken instructions and dialogue. Such failures can therefore reflect missing input rather than insufficient reasoning or an inadequate action space.

Prior work enriches individual screenshots (Zhou et al. 2023; Deng et al. 2023; Yang et al. 2023; Zheng et al. 2024), evaluates temporal reasoning in video environments (Chen et al. 2024; Jang et al. 2024; Zhang et al. 2025; Chen et al. 2025), or processes continuous streams (Google 2025; OpenAI 2024). These directions leave a distinct question: can an existing screenshot-based CU agent acquire and retain multimodal information between actions without retraining? We use *observation interface* for the sampling time, modalities, representation, and retention of information supplied to an agent. Following work on action interfaces (Yang et al. 2024), we treat observation as independently configurable from the model.

We introduce the **Agent-Computer Observation Interface (AOI)**, a layer between the environment and an image-and-text CU agent. Between decisions, a pixel-change gate retains informative keyframes and a volume gate triggers speech transcription. The CU model also narrates new visual information, which persists as text after older images are removed. AOI supplies these channels with the current screenshot in a structured record without changing the model parameters or action space.

We evaluate AOI using **DynaCU-Bench**, which contains 100 dynamic browser tasks across 10 families, together with a 50-task static control. The main evaluation covers six closed- and open-source CU models, with two Qwen3-VL models as an independent replication. AOI improves all eight by 17–48 percentage points without retraining. It completes 50/50 static-control tasks versus 43/50 for the screenshot baseline. On a 12-task audio subset, AOI-equipped Claude completes all tasks, versus 3 for OpenAI Realtime and 0 for Gemini Live.

Ablations show that both the structured record and new perceptual input contribute to the gains, that keyframes provide most of their value through narration, and that the best channel combination is model-dependent.

Contributions.

1. We formulate the observation interface through sampling time, modality, representation, and retention, and develop AOI to decouple observation from action.
2. We introduce DynaCU-Bench: 100 dynamic browser tasks across 10 families and a 50-task static control.
3. Across eight CU models and two streaming baselines, we identify narration and model-specific channel selection as key factors (Sections 5–7).

2 Background: The CU Agent Loop

Every current CU agent follows a fixed *observe → reason → act* cycle: capture a screenshot s_t, sample an action a_t conditioned on s_t and trajectory history τ, execute it, wait a short buffer δ, and repeat. Two spaces define an agent's capabilities. The **action space** $\mathcal{A}$ is a small grammar of discrete commands—click, type, scroll, wait, done—that has seen substantial prior innovation (Yang et al. 2024). By contrast, the **observation space** $\mathcal{S}$ remains severely limited to a single RGB screenshot per step.

This design leaves agents blind between screenshots. The typical 3–5 second interval means motion, and transient UI events are almost entirely missed. Furthermore, the lack of an audio channel leaves agents deaf to all sounds. Concretely, four categories of content are systematically missed: (i) *transient events*—toasts and dialogs that appear and auto-dismiss within a single interval; (ii) *continuous visual media*—video, animated tutorials, and transitioning slides; (iii) *audio*—meeting speech, notification sounds, and spoken prompts; and (iv) *periodic noise*—spinners and blinking cursors that should be *ignored* yet disrupt naive frame differencing methods. AOI expands $\mathcal{S}$ to cover (i)–(iii) while suppressing (iv), all without modifying the model or $\mathcal{A}$.

3 System Design

The AOI is a lightweight, model-agnostic perception layer inserted between the environment and any existing CU model. It continuously observes the full interval between agent steps and supplies additional keyframes, audio transcripts, and visual narrations in standard image+text format. On static, silent tasks it produces almost no overhead and falls back to the standard loop wrapped in a structured observation record (which slightly improves static performance). On dynamic tasks the model receives inter-step keyframes, audio, and persistent narration. No retraining is required, though the optimal component mix is model-specific. The added observation work increases latency on fast models, so AOI targets human-paced rather than real-time use.

3.1 Architecture Overview

AOI has three gated components (Figure 1), each preceded by a fast (sub-millisecond) gate that skips processing on static or silent content. Additional cost when gates fire is dominated by CLIP-ViT-B/16 (∼7 ms) for visuals and Whisper large-v3 for audio.

(1) **Inter-step keyframe capture**: Samples the screen at ∼3 Hz and selects up to 5 keyframes per step using simple pixel-change gating (CLIP-based semantic filtering is optional).

(2) **Volume-gated audio observer**: RMS energy gate followed by Whisper large-v3 transcription, active only when speech is detected.

(3) **Visual narration context**: The CU model itself generates a brief description of new visual content as a side-output of each inference call. Narrations persist as text after images are pruned.

Algorithm 1: Two-stage adaptive keyframe extraction (CLIP variant). This algorithm is shown for completeness; the default selector in every main result is the simpler pixel-only gate (Stage 1).

Require: Screen sample x_t, anchor embedding e_{anchor}, pixel threshold α, CLIP threshold θ

1: $\Delta_{\text{px}} \leftarrow \text{PixelChangeRatio}(x_t, x_{\text{prev}})$
2: **if** $\Delta_{\text{px}} < \alpha$ **then**
3:     **return** Skip {Stage 1: no pixel change}
4: **end if**
5: $e_t \leftarrow \text{CLIP}(x_t)$
6: $d \leftarrow 1 - \cos(e_t, e_{\text{anchor}})$
7: **if** $d > \theta$ **then**
8:     $e_{\text{anchor}} \leftarrow e_t$ {Re-anchor}
9:     **return** CaptureKeyframe(x_t, t)
10: **end if**
11: **return** Skip {Stage 2: below semantic threshold}

3.2 Inter-Step Keyframe Capture

Multiple selection strategies are possible—uniform fixed-rate sampling, pixel-level differencing, random sampling, or semantic filtering via CLIP embeddings (Radford et al. 2021). We use simple pixel-change gating for its low overhead: if fewer than 1% of pixels changed since the last captured frame, the sample is skipped (cost: <1 ms on CPU). Algorithm 1 formalises the two-stage CLIP variant; the default selector drops the CLIP stage (lines 5–10) and uses only Stage 1.

3.3 Volume-Gated Audio Observation

When audio is present, we transcribe speech with Whisper large-v3 (Radford et al. 2023), operating on 16 kHz mono audio captured from the system audio output via PulseAudio virtual devices. Our implementation focuses on transcribing *speech* only; non-speech audio events may trip the volume gate but will produce no useful transcript. The audio buffer spans the full inter-step interval, including the post-action buffer. This ensures speech that begins just after an action (e.g. a spoken confirmation) is still captured. The audio buffer also includes ∼3.5 seconds of overlap from the previous interval to provide acoustic continuity across step boundaries.

3.4 Visual Narration for Long-Term Context

CU models retain only the most recent images in context, pruning earlier keyframes. Consequently, on tasks that span many steps, the agent loses all visual memory of prior content.

To address this, the CU model also generates brief visual narration in the same inference call that produces the action and persists indefinitely in the trajectory, even after the corresponding images are pruned. Following the caption-then-reason principle of LLoVi (Zhang et al. 2023), narrations are generated by the CU model itself and are task-relevant rather than generic.

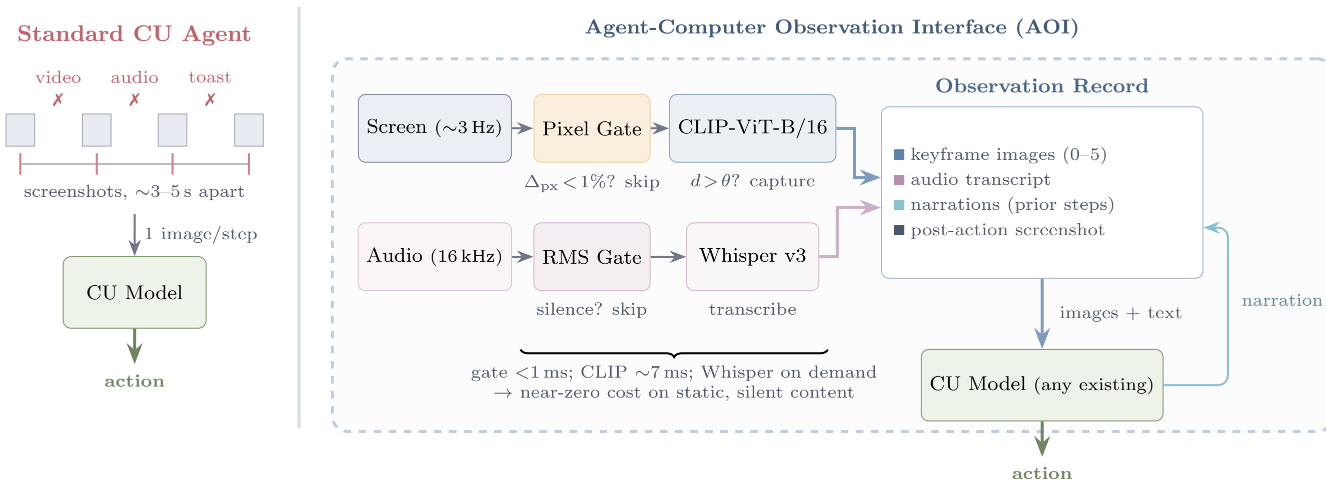

Figure 1: **Standard CU agents vs. AOI-equipped agents.** *Left:* Standard agents observe only periodic screenshots and therefore miss audio and inter-step visual events. *Right:* AOI continuously monitors screen and audio streams, extracts only informative observations, and appends them as persistent text to the observation record before invoking the underlying CU model.

Observation Record — Step N (Meeting task)

```
# CONTEXT -- text from prior steps (no images)
Step N-1 (t ≈ 18.5-22.0 s):
AUDIO: "Here's the team. As you can see we
have strong cross-functional coverage."
VISUAL: "Meeting slide shows Project Team
with 6 members listed in a grid."
ACTION: wait()

# NEW -- current interval observations
Step N (t ≈ 22.0-25.5 s):
AUDIO: "And I want to mention, we've moved
the launch date to April 28th.
Please update your calendars."
[IMAGE: post-action screenshot]

# TASK
Read the meeting slides and listen to the
discussion. What is the new launch date?
Enter it in the Launch Date field.
```

Figure 2: **Example observation record at step N of a meeting task.** Prior observations persist as text, while the current step contributes a fresh audio transcription and screenshot. In this example, no keyframes are included because the slide does not change.

3.5 Observation Record

At each step the model receives a structured record combining persistent text context (prior audio, narrations, actions) with current observations (new audio transcription, keyframes if any, and post-action screenshot). On static tasks this reduces to the screenshot plus structured scaffold. Figure 2 shows an example.

3.6 Implementation

AOI is implemented as a modular ∼2,600-line Python wrapper. Screen and audio are captured via headless browser and virtual audio devices. Keyframe processing runs on GPU while ASR runs on CPU to avoid contention. Cloud models are called through their native APIs and local models are served on a single GPU.

4 DynaCU-Bench

4.1 Design Principles

Existing CU benchmarks (Xie et al. 2024; Zhou et al. 2023; He et al. 2024) focus almost exclusively on static tasks. We introduce **DynaCU-Bench**, a 100-task benchmark of realistic browser-based tasks that require dynamic visual or audio perception.

The benchmark follows five design principles: tasks must be unsolvable from static screenshots, reflect real computer use, run reproducibly in a headless browser, cover audio, visual-temporal, and interaction axes, and include difficulty stratification.

It spans 10 task families (Table 1). Each family contains 10 tasks with balanced difficulty (3 easy, 4 medium, 3 hard).

4.2 Evaluation Protocol

Each task runs in a headless Chromium browser with virtual audio devices for playback and microphone injection; spoken content is rendered using high-quality text-to-speech. Most tasks (93) are scored by a deterministic check on the resulting page state. The remaining tasks either combine that check with an LLM quality rubric (6) or rely solely on the LLM judge (1). A run ends after 15 agent steps or a per-task wall-clock budget (the stimulus duration plus a fixed margin and any AOI overhead), whichever occurs first. These budgets bind only for the slowest models (e.g. Grok-4 at ∼80–180 s/call; Section 7).

Table 1: DynaCU-Bench task families. Axes: AUD = audio perception, VIS = visual-temporal perception, INT = real-time interaction. Each family contains 3 easy, 4 medium, and 3 hard tasks.

Family	Axes	Description
Podcast	AUD	Listen to spoken narration; extract facts or answer questions
Meeting	AUD+VIS+INT	Follow a meeting with slides and speech; take notes or act
Screencast	VIS	Watch terminal/IDE recordings with auto-advancing frames
Carousel	VIS	Perceive CSS transitions, rotating content, product carousels
Dashboard	VIS	Monitor live-updating dashboards
Transient	VIS	Respond to toasts and auto-dismissing banners
Phone	AUD+INT	Listen to a synthesized voice call and respond
Interview	AUD+INT	Answer spoken questions; complete interviews
Collab	VIS+INT	Handle remote edits in collaborative documents
Games	VIS+INT	Play interactive games requiring temporal attention

5 Evaluation

5.1 Setup

CU models. We evaluate six computer-use models spanning closed- and open-source systems: Claude Sonnet 4.6 (Anthropic 2024a), GPT-5.4 (OpenAI 2026), Gemini 2.5 Flash (Google DeepMind 2025), Grok-4 (xAI 2026), EvoCUA-32B (Meituan 2026), and Fara-7B (Awadallah et al. 2025). We additionally evaluate Qwen3-VL-235B-A22B-Instruct and Qwen3-VL-30B-A3B-Instruct (Qwen Team 2025) via OpenRouter as an independent open-source replication (Table 2). All models use vendor-recommended decoding settings and are evaluated directly in our DynaCU-Bench harness.

Observation configurations. The primary contrast is between **Standard** (one screenshot per step, no audio—the current paradigm) and **AOI full** (inter-step keyframes + audio transcription + visual narration). We evaluate both modes across all six models. All remaining modes are ablations and diagnostic controls run on Claude Sonnet 4.6 (unless otherwise noted).

5.2 Main Results

Figure 3 shows the main results. AOI substantially improves performance across all six models. For example, on meeting tasks the standard agent often fails because it cannot hear spoken answers, while AOI transcribes the audio and completes the task in far fewer steps. On video and carousel tasks, narration often enables correct answers in a single step.

Statistical protocol. Each configuration is evaluated once on 100 binary tasks. We report paired McNemar exact mid-p tests. Independent seed sweeps later show approximately 3 pp decoding variability, so small differences should not be over-interpreted.

Figure 3 shows that AOI improves task success for every model, with gains ranging from +17 to +48 percentage points. Claude Sonnet 4.6 achieves the highest absolute score (82%), while Gemini 2.5 Flash shows the largest gain (+48 pp). Open-source models also benefit substantially: EvoCUA-32B improves from 18% to 55%, and even the 7B Fara model doubles its success rate (17%→34%). These consistent gains indicate that AOI addresses a perception bottleneck rather than exploiting model-specific behavior.

Additional open-source replication (Qwen3-VL family). Both Qwen3-VL models reproduce the same trend: the 235B model improves by +42 pp (22%→64%) and the 30B model by +24 pp (18%→42%), reinforcing that AOI's gains are model-agnostic.

Table 2: Open-source replication on DynaCU-Bench (100 tasks) using Qwen3-VL models. Results use the same evaluation protocol as Figure 3.

Model	Standard	AOI full	Δ
235B-A22B	22	**64**	+42
30B-A3B	18	**42**	+24

Variance. Repeating the headline configuration (Claude + AOI full) across three seeds yields 82%, 78%, 76% (mean 78.7%, std 3.1 pp). The reported 82% thus sits at the upper end of a $\sim$3 pp decoding variability band. This quantifies absolute score variance and is complementary to the paired McNemar tests reported above.

5.3 Per-Family Analysis

Figure 4 reveals three patterns. Audio tasks (Podcast, Phone, Meeting) improve the most because they are impossible from screenshots alone. Visual-temporal tasks (Screencast, Carousel, Transient) also improve consistently, whereas Dashboard gains are modest because the relevant information is often visible in a single screenshot. Games is the only family that consistently regresses: it is limited by action latency rather than perception, and AOI's additional observation time exceeds its benefit.

5.4 Difficulty Breakdown

AOI improves performance across all difficulty tiers, with the largest absolute gains on medium and hard tasks for stronger models. On Fara-7B, improvements are concentrated on easy tasks, suggesting that reasoning rather than perception is the dominant remaining bottleneck.

5.5 Efficiency

Better perception reduces the number of interaction steps enough to outweigh the richer observations, lowering token costs by 15–50% across cloud models. Wall-clock time decreases only for the slowest models, where fewer agent

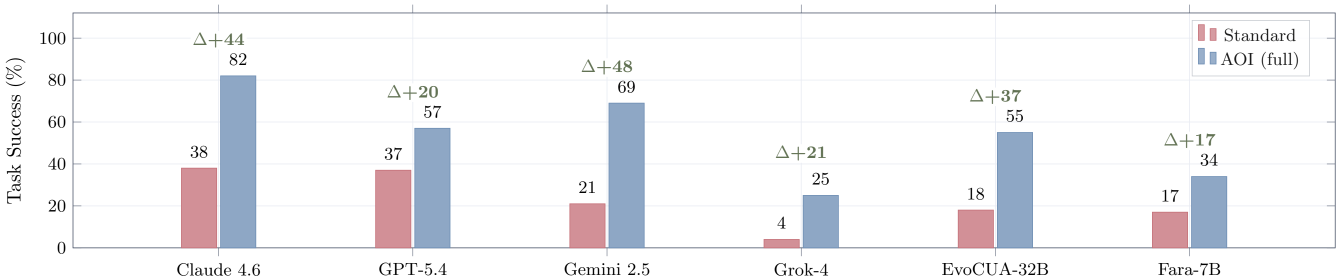

Figure 3: **Main results on DynaCU-Bench (100 tasks).** AOI substantially improves task success across all evaluated CU models without retraining. Green labels indicate the absolute improvement over the standard screenshot baseline.

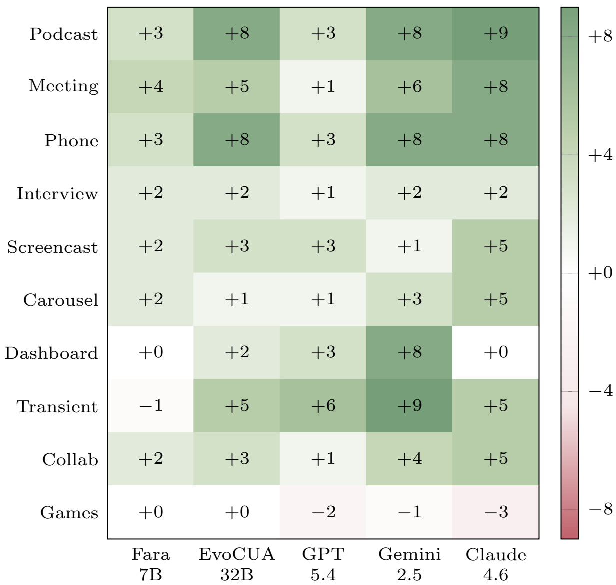

Figure 4: **Per-family AOI gain** (AOI full − Standard, tasks per 10) across five models. Families are grouped by modality. Green indicates improvement; red indicates regression.

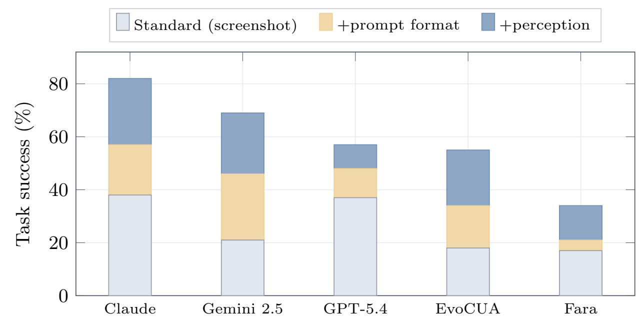

Figure 5: Bars stack the standard screenshot baseline, the gain from the structured observation record alone, and the additional gain from perception (keyframes, audio, and narration).

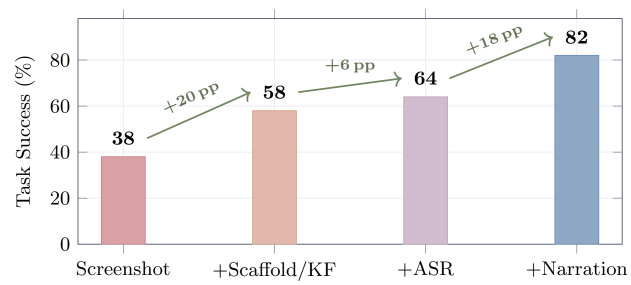

Figure 6: Progressive contribution of AOI components on Claude Sonnet 4.6.

steps dominate the added observation overhead. AOI therefore improves cost efficiency while trading off latency on faster models.

6 Decomposing the Gain

We decompose AOI's gains in two stages: separating the structured prompt-format effect from genuine perception improvements, then breaking down the perception component channel by channel.

6.1 Prompt Format vs. Perception

We first rule out the possibility that AOI helps only through prompt reformatting.

On the purely static **Static-50** benchmark, AOI achieves a perfect 50/50 while the screenshot baseline scores 43/50, demonstrating a beneficial prompt-format effect with no degradation.

On dynamic tasks, a control using only AOI's structured observation record (no keyframes or audio) isolates the prompt-format contribution. Figure 5 shows both prompt-format and perception components are positive for every model. The balance is model-specific: Claude is perception-heavy, Gemini 2.5 is prompt-heavy, and Fara-7B cannot exploit the richer prompt. Within the prompt component, the DOM-element list is the largest lever.

6.2 Perception Channel Contributions

Figure 6 shows incremental gains. Keyframe images add only +1 pp as raw input but +10 pp once narrated (Figure 7b), with benefits concentrated on Transient and Carousel tasks. Table 3 reports results across keyframe selection strategies. Audio transcription adds directional gains on audio families, while visual narration is the largest single component (+18 pp).

Table 3: Results for successive ablation configurations on Claude Sonnet 4.6. Total reports successful tasks out of 100.

Comparison	Total
Standard → Uniform 1 FPS	38 / 58
Uniform 1 FPS → Uniform 3 FPS	58 / 56
Uniform 3 FPS → Pixel-diff	56 / 58
Pixel-diff → Random keyframes	58 / 56
Random keyframes → CLIP adaptive	56 / 58
CLIP adaptive → AOI visual+ASR	58 / 64
AOI visual+ASR → AOI full	64 / 82

A narration-discarded control (Figure 7a) attributes most of narration's benefit to inference-time articulation, with persistence mattering mainly on multi-step tasks.

The keyframe channel is the most model-dependent: strongly positive with narration on Claude and Gemini 2.5, but regressing on Gemini 3 due to image-token dilution. Overall, AOI components must be tuned per model.

7 Model Sensitivity Study

To characterize sensitivity to the underlying model and its inference latency, we evaluate AOI on three additional model configurations (Table 4).

Grok models are primarily limited by high inference latency rather than perception. Grok-4 achieves only +21 pp with AOI due to frequent timeouts. The Grok-4-fast-reasoning and Grok-4.3 configurations achieve larger gains (+28 pp and +40 pp, respectively), indicating that reduced latency allows more of AOI's perceptual benefit to be realized within the task budget.

Gemini 3 Flash achieves only a modest +9 pp net gain. A four-way decomposition (Figure 8) reveals a cancellation of effects: audio and the structured scaffold help as usual, but the keyframe stream regresses by -12 pp due to image-token dilution. Dropping keyframes alone recovers +21 pp over the standard baseline, showing the default AOI bundle is mis-tuned for this model.

These cases illustrate that AOI gains are robust but model-specific: the optimal component mix must be chosen per model rather than applied as a fixed bundle.

8 Comparison with Streaming Multimodal Baselines

A natural alternative is a *streaming multimodal model* that integrates perception and reasoning, such as Gemini Live (Google 2025) or the OpenAI Realtime API (OpenAI 2024). Neither is a computer-use agent, so we adapt both to DynaCU-Bench by streaming page audio and screenshots into a single session and executing returned tool calls. We evaluate the highest vision- and function-calling-capable tier of each on the same 12-task audio subset used for AOI.

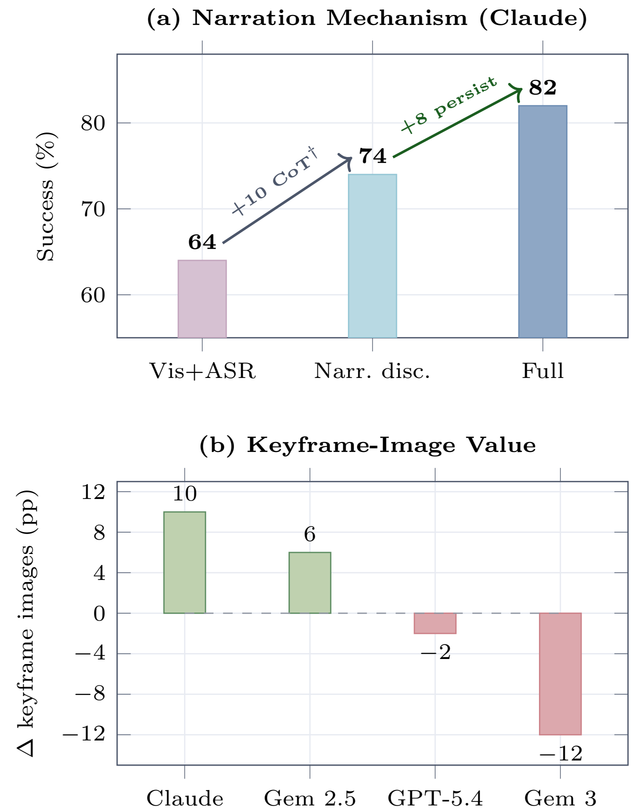

Figure 7: **Contribution of narration and keyframes. (a)** Visual narration improves performance, with most of the gain arising from inference-time articulation rather than persistent memory. **(b)** The value of keyframes is model-dependent: positive for Claude and Gemini 2.5, neutral for GPT-5.4, and negative for Gemini 3.

Both underperform the AOI by a wide margin (Table 5). OpenAI Realtime succeeds only on interview questions answerable from a single audio chunk, while Gemini Live rarely emits usable tool calls. A sanity check on five purely visual tasks confirms the adapters function correctly: both baselines execute valid browser actions (5/5) but solve none of the tasks, whereas AOI-equipped Claude solves 4/5. Streaming multimodal interaction alone is therefore insufficient; the critical capability is persisting transient observations as text that can guide future actions.

9 Related Work

Interface Design over Model Training. Much agent progress comes from redesigning the agent–computer interface rather than the model weights. Action interfaces have evolved through richer command sets (Yang et al. 2024), executable code actions (Wang et al. 2024a), and element-grounded actions (Deng et al. 2023). Observation interfaces have instead focused on enriching individual screenshots with accessibility trees, HTML surrogates, and overlays (Zhou et al. 2023; Deng et al. 2023; Yang et al. 2023;

Table 4: Model-sensitivity results on DynaCU-Bench (100 tasks per configuration). Gemini 3 Flash probes sensitivity to observation composition, while the Grok configurations probe sensitivity to inference latency.

Model	Standard	AOI-full	Δ
Gemini 3 Flash	36	45	+9
Grok-4.3	25	65	+40
Grok-4-fast-reasoning	19	47	+28

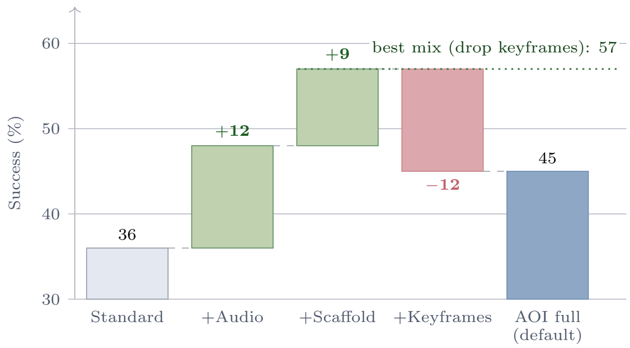

Figure 8: **Gemini 3 Flash achieves only a small net gain (+9 pp).** Audio and the structured scaffold remain beneficial, but keyframes reduce performance by 12 pp.

Zheng et al. 2024). None addresses temporal or multimodal observation.

Computer-Use Agents. Recent CU agents span closed-source systems (Anthropic 2024a; OpenAI 2025; Google DeepMind 2025; Andreux et al. 2025) and open-source models (Wang et al. 2025a,b; Awadallah et al. 2025; Ye et al. 2025; Song et al. 2025), with Agent S3 (Simular AI 2025) demonstrates competitive open-source performance. Existing surveys (Sager et al. 2025) discuss many challenges but not observation as a distinct design axis; all current agents retain the screenshot-per-step loop.

Streaming Multimodal Models. Gemini Live (Google 2025) and OpenAI Realtime (OpenAI 2024) process continuous streams but bundle perception and reasoning in one model. Section 8 shows they underperform AOI on grounded computer-use tasks. Our work is complementary: AOI provides a modular perception layer usable by any CU model.

Video and GUI Understanding. Benchmarks such as GUI-World (Chen et al. 2024), VideoWebArena (Jang et al. 2024), CUA-Suite (Chen et al. 2025), and MemGUI-Bench (Park et al. 2025) expose the limitations of static observation and GUI memory, but do not provide observation mechanisms compatible with existing CU agents.

Keyframe selection. Prior work studies CLIP-based frame selection and sampling for offline video QA (Wang et al. 2024b; Tang et al. 2025; Zohar et al. 2024). In contrast, AOI performs real-time observation filtering for interactive computer use, combining a sub-millisecond pixel gate with CLIP only on changed frames.

Adaptive middleware for GUI agents. Cradle (Tan et al.

Table 5: Streaming multimodal baselines vs. AOI on a 12-task audio subset (3 tasks each from Podcast, Meeting, Phone, and Interview).

Configuration	Pod	Meet	Phone	Intv	Total / 12
OpenAI Realtime	0	0	0	3	3 (25%)
Gemini Live	0	0	0	0	0 (0%)
AOI full (Claude 4.6)	3	3	3	3	**12** (100%)

2024), ScreenLLM (Jin et al. 2025), StreamAgent (Yang et al. 2025), and Dispider (Qian et al. 2025) separate or adapt perception, but none combines adaptive observation, audio, and persistent multimodal memory for computer-use agents.

Structured agent interfaces. Protocols such as MCP (Anthropic 2024b) and NLWeb (Microsoft 2025) bypass perception through structured APIs. They require website cooperation, whereas AOI operates on arbitrary unmodified interfaces.

10 Limitations

Latency and temporal resolution. AOI extends perception but not decision speed: CU models still require 1–5 s per step and the screen is sampled at $\sim$3 Hz, placing sub-$\sim$300 ms events and real-time games outside its scope.

Perception is not reasoning. Better observations do not improve reasoning; on smaller models, reasoning capacity remains the primary bottleneck.

Audio is speech-only and narration can err. AOI transcribes speech but not non-speech sounds, and visual narrations can propagate recognition errors through the trajectory.

External validity. Experiments use a controlled browser with synthesized audio. Evaluating AOI on existing dynamic-GUI benchmarks requires re-instrumenting each benchmark's interaction harness and remains future work.

Statistical power. Each configuration is evaluated once on 100 tasks. Paired McNemar tests support comparisons, but differences of only a few percentage points should not be over-interpreted.

11 Conclusion

Computer-use agents are blind between screenshots and deaf to audio because observation is tied to action: one screenshot per step. We argued the *observation* is a separable design axis and introduced the AOI, a perception layer that is transparent on static work, adaptive on dynamic content, and compatible with existing CU models without retraining. Across eight models, AOI turns many dynamic tasks that are nearly impossible from screenshots into solvable ones while reducing token cost.

Our analysis further shows that keyframe selection alone contributes little: gains emerge when observations are converted into persistent text, and the best observation recipe is model-dependent rather than universal. We release the AOI and DynaCU-Bench to support future research on observation interfaces for computer-use agents.

References

Andreux, M.; et al. 2025. Surfer 2: The Next Generation of Cross-Platform Computer Use Agents. *arXiv preprint arXiv:2510.19949*.

Anthropic. 2024a. Claude Computer Use. https://docs.anthropic.com/en/docs/agents-and-tools/computer-use.

Anthropic. 2024b. Model Context Protocol. https://modelcontextprotocol.io.

Awadallah, A.; et al. 2025. Fara-7B: An Efficient Agentic Model for Computer Use. *arXiv preprint arXiv:2511.19663*.

Chen, D.; et al. 2024. GUI-World: A Video Benchmark and Dataset for Multimodal GUI-oriented Understanding. *arXiv preprint arXiv:2406.10819*.

Chen, L.; et al. 2025. CUA-Suite: 55 Hours of Real Computer-Use Recordings for Agent Benchmarking. *arXiv preprint arXiv:2510.10142*.

Deng, X.; Gu, Y.; Zheng, B.; Chen, S.; Stevens, S.; Wang, B.; Sun, H.; and Su, Y. 2023. Mind2Web: Towards a Generalist Agent for the Web. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Google. 2025. Live API (Gemini Live). https://ai.google.dev/gemini-api/docs/live.

Google DeepMind. 2025. Gemini 2.5 Computer Use. https://deepmind.google/models/gemini/computer-use/.

He, H.; et al. 2024. WebVoyager: Building an End-to-End Web Agent with Large Multimodal Models. *arXiv preprint arXiv:2401.13919*.

Jang, L.; et al. 2024. VideoWebArena: Evaluating Long Context Multimodal Agents with Video Understanding Web Tasks. *arXiv preprint arXiv:2410.19100*.

Jin, Y.; et al. 2025. ScreenLLM: Stateful Screen Schema for Efficient Action Understanding and Prediction. *arXiv preprint arXiv:2503.20978*.

Meituan. 2026. EvoCUA: Evolving Computer Use Agent. https://github.com/meituan/EvoCUA. Accessed 2026-05-14.

Microsoft. 2025. NLWeb: A Conversational Interface for Websites. https://github.com/microsoft/NLWeb.

OpenAI. 2024. Realtime API: Speech-to-speech models with tool use. https://platform.openai.com/docs/guides/realtime.

OpenAI. 2025. Computer-Using Agent (CUA). https://openai.com/index/computer-using-agent/.

OpenAI. 2026. Introducing GPT-5.4. https://openai.com/index/introducing-gpt-5-4/.

Park, J.; et al. 2025. MemGUI-Bench: Benchmarking Memory in GUI Agents. *arXiv preprint arXiv:2509.12233*.

Qian, R.; et al. 2025. Dispider: Enabling Video LLMs with Active Real-Time Interaction via Disentangled Perception, Decision, and Reaction. *arXiv preprint arXiv:2501.03218*.

Qwen Team. 2025. Qwen3-VL Technical Report. *arXiv preprint arXiv:2511.21631*.

Radford, A.; Kim, J. W.; Xu, T.; Brockman, G.; McLeavey, C.; and Sutskever, I. 2023. Robust Speech Recognition via Large-Scale Weak Supervision. In *ICML*. ArXiv:2212.04356.

Radford, A.; et al. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *ICML*. ArXiv:2103.00020.

Sager, P. J.; et al. 2025. A Comprehensive Survey of Agents for Computer Use: Foundations, Challenges, and Future Directions. *arXiv preprint arXiv:2501.16150*.

Simular AI. 2025. Agent S3: An Open-Source Computer-Use Agent. https://github.com/simular-ai/Agent-S.

Song, L.; et al. 2025. CoAct-1: Computer-using Agents with Coding as Actions. *arXiv preprint arXiv:2508.03923*.

Tan, W.; et al. 2024. Cradle: Empowering Foundation Agents Towards General Computer Control. *arXiv preprint arXiv:2403.03186*.

Tang, X.; et al. 2025. Adaptive Keyframe Sampling for Long Video Understanding. *arXiv preprint arXiv:2502.21271*.

Wang, H.; et al. 2025a. UI-TARS-2 Technical Report: Advancing GUI Agent with Multi-Turn Reinforcement Learning. *arXiv preprint arXiv:2509.02544*.

Wang, X.; Chen, Y.; Yuan, L.; Zhang, Y.; Li, Y.; Peng, H.; and Ji, H. 2024a. Executable Code Actions Elicit Better LLM Agents. In *International Conference on Machine Learning (ICML)*.

Wang, X.; et al. 2025b. OpenCUA: Open Foundations for Computer-Use Agents. *arXiv preprint arXiv:2508.09123*.

Wang, Z.; et al. 2024b. VideoTree: Adaptive Tree-based Video Representation for LLM Reasoning on Long Videos. *arXiv preprint arXiv:2405.19209*.

xAI. 2026. Grok 4. https://x.ai/news/grok-4. Accessed 2026-05-14.

Xie, T.; et al. 2024. OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments. *arXiv preprint arXiv:2404.07972*. NeurIPS 2024.

Yang, H.; et al. 2025. StreamAgent: Towards Anticipatory Agents for Streaming Video Understanding. *arXiv preprint arXiv:2508.01875*.

Yang, J.; Zhang, H.; Li, F.; Zou, X.; Li, C.; and Gao, J. 2023. Set-of-Mark Prompting Unleashes Extraordinary Visual Grounding in GPT-4V. *arXiv preprint arXiv:2310.11441*.

Yang, J.; et al. 2024. SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering. *arXiv preprint arXiv:2405.15793*.

Ye, J.; et al. 2025. Mobile-Agent-v3: Fundamental Agents for GUI Automation. *arXiv preprint arXiv:2508.15144*.

Zhang, A. L.; et al. 2025. VideoGameBench: Can Vision-Language Models complete popular video games? *arXiv preprint arXiv:2505.18134*.

Zhang, C.; et al. 2023. A Simple LLM Framework for Long-Range Video Question-Answering. *arXiv preprint arXiv:2312.17235*.

Zheng, B.; Gou, B.; Kil, J.; Sun, H.; and Su, Y. 2024. GPT-4V(ision) is a Generalist Web Agent, if Grounded. In *International Conference on Machine Learning (ICML)*.

Zhou, S.; et al. 2023. WebArena: A Realistic Web Environment for Building Autonomous Agents. *arXiv preprint arXiv:2307.13854*. ICLR 2024.

Zohar, O.; et al. 2024. Apollo: An Exploration of Video Understanding in Large Multimodal Models. *arXiv preprint arXiv:2412.10360*.